“In The Name Of God”

Assignment 3: Aspect-Based Hybrid Recommender Systems via Rule-Based Text Mining on Yelp Reviews

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1. Problem Setting:

Why ratings alone fail to capture user preferences and The importance of review text mining:

Rating signals alone are too coarse to fully represent user preferences in restaurant recommendation. Two restaurants may receive the same average rating while eliciting very different reactions from different users. One user may care most about food quality, another about service speed, and a third about ambience or price. A scalar rating collapses all of these distinctions into a single number, which makes it difficult to learn the reasons behind preference and to generalize across sparse users or sparse items.

Review text provides the missing structure. In restaurant reviews, users often mention concrete aspects such as food, service, ambience, wait time, and price. Mining these reviews makes it possible to infer aspect-level preferences and aspect-level item quality. This is especially important in sparse recommendation settings, where many user-item pairs have few or no observed ratings. In such cases, textual evidence helps compensate for the lack of interaction history and improves both interpretability and coverage.

2. Method Overview:

End-to-end architecture diagram from raw CSV/JSONL to final hybrid recommendations:

The proposed system is a hybrid recommender that combines collaborative filtering with review-text understanding. The pipeline starts from raw CSV and JSONL files and ends with ranked restaurant recommendations and trace logs for explanation.

The overall workflow is:

1. Load raw review and business metadata.

2. Clean and preprocess text.
3. Extract aspect mentions from reviews.
4. Map surface forms to canonical aspects using semantic similarity.
5. Perform dependency-guided sentiment analysis on aspect mentions.
6. Build user aspect profiles and item aspect profiles.
7. Train or load an SVD-based rating predictor.
8. Normalize rating-based and content-based scores.
9. Blend both signals using a weighting parameter alpha.
10. Generate top-K recommendations and write trace.jsonl.
11. Evaluate and compare to other baselines

A compact architecture view is:

raw CSV/JSONL -> preprocessing -> aspect extraction -> sentiment profiling -> SVD prediction -> score normalization -> hybrid fusion -> ranked recommendations -> Evaluate and run other baselines -> (trace.jsonl, summery.json, dev_metrics.json)

3. Aspect Extraction & Semantic Mapping:

Detail POS tagging rules, seed aspects similarity thresholds, and sample mapping outcomes:

Aspect extraction identifies opinion-bearing nouns and noun phrases from review text. The system uses POS-based patterns and seed-aspect matching to detect candidates such as food, service, price, ambience, and location.

A practical extraction strategy is:

- noun tokens and noun phrases are collected as aspect candidates
- adjective-noun pairs (root tokens) are inspected for aspect-bearing phrases
- pre-defined seed aspects guide canonicalization
- semantically similar phrases are mapped to the closest canonical aspect when similarity exceeds a threshold

For example:

- waiter, staff, and server can map to service
- meal, dish, and burger can map to food
- atmosphere, decor, and setting can map to ambience

This semantic mapping reduces sparsity. Without it, the same concept would appear under many surface forms and fragment the user profile. A similarity threshold is used to avoid over-merging unrelated terms. If the threshold is too low, false mappings increase; if it is too high, useful paraphrases are missed. The chosen threshold therefore balances precision and recall in aspect canonicalization.

4. Dependency-Guided Sentiment Analysis:

Detailed tree-traversal logic (e.g. amod, copula structures), negation checking rules, and user/item profile distributions:

For each matched aspect mention, the system does not score the entire sentence. Instead, it uses spaCy dependency parsing to extract opinion expressions attached to the aspect head.

The implementation follows only two required patterns:

1. Adjectival modifier (amod)

   If the aspect head has adjectival children with dependency label amod, those adjectives are extracted.

   - Example: "We had extremely friendly service."
   - Aspect Head: “service”
   - Adjectival child: “friendly”
   - Friendly’s advmod: “extremely”
   - Extracted: “extremely friendly”

2. Copular adjectival complement

   If the aspect head is an nsubj or nsubjpass of a copular construction, the adjectival complement of the copular verb is extracted.

   - Example: "The food was not very good."
   - Aspect Head: “food”
   - Acomp for “was”: “good”
   - Good’s advmod: “very”

- Neg: “not”
- Extracted: “not very good”

For every extracted adjective, the system includes its contiguous advmod and neg children, sorts the selected tokens by token index, and joins their surface forms with spaces. Thus phrases such as ‘extremely clean’ and ‘not good are valid’ extracted opinion phrases.

Each extracted phrase is scored with the **standard NLTK VADER SentimentIntensityAnalyzer**, using the compound value. For a review $r$and canonical aspect a, the aspect sentiment score is the arithmetic mean of all phrase scores mapped to that aspect:

$$s(r,a) = \frac{1}{|\,\mathcal{P}_{r,a}\,|} \sum_{p \in \mathcal{P}_{r,a}} \text{VADER}\,(p)$$

If an aspect is mentioned but no opinion phrase is extracted, then:

$$s(r,a) = 0.0$$

Reviews with no mention of aspect $a$are excluded from $\mathcal{R}_i^{(a)}$.

These scores are then aggregated into two profile matrices:

User-Aspect Attention Matrix U

$$U_{u,a} = \frac{\text{count}(u,a)}{\max\limits_{a'} \text{count}(u,a'), \epsilon}$$

where count(u,a) is the number of mapped mentions of aspect a in user u’s training reviews, and $\epsilon = 10^{-9}$.

Each row of U sums to 1 if the user mentions at least one aspect; otherwise it is all zeros.

Item-Aspect Quality Matrix V

$$V_{i,a} = \frac{\sum_{r \in \mathcal{R}_i^{(a)}} s\,(r,a)}{\max(\text{count}, (i,a)\epsilon)}$$

where count(i,a) is the number of reviews of item i that mention aspect a.

Each entry lies in [-1,1].

5. Hybrid Recommendation Formulation:

Equations for SVD matrix factorization, aspect weighted alignment, global score normalization, and blending:

The final prediction for user u and business i is computed as a weighted combination of collaborative filtering and aspect-based sentiment alignment:

$$\hat{R}_{u,i} = \alpha \cdot ScoreCF_{\text{norm}}(u,i) + (1-\alpha) \cdot ScoreAspect_{\text{norm}}(u,i)$$

where $\alpha \in [0,1]$ controls the trade-off between rating-based preference modeling and review-text-based personalization.

1. Collaborative Filtering Component

We train a classical biased matrix factorization model on the explicit rating matrix $R \in \mathbb{R}^{|U|\times|I|}$. If multiple training reviews exist for the same (u,i)pair, their arithmetic mean is used as the observed rating. The prediction is:

$$ScoreCF(u,i) = \mu + b_u + b_i + p_u^{\top} q_i$$

where:

- μ is the global mean rating,
- b_u and $b_i$are user and item bias terms,
- $p_u, q_i \in \mathbb{R}^k$are latent user/item vectors,
- $k$is the latent dimension.

The model is trained only on training reviews. All random sources are fixed using a seed for reproducibility.

2. Aspect Sentiment Alignment Component

To combine user aspect preferences with item aspect quality, we compute:

$$ScoreAspect(u,i) = \sum_{a=1}^{5} U_{u,a} \cdot V_{i,a}$$

where $U_{u,a} \geq 0$ is the normalized user attention weight for aspect a, and $V_{i,a} \in [-,1+1]$is the sentiment quality score of item $i$on aspect a.

Using non-negative user weights ensures that negative item sentiment on an important aspect decreases the score instead of producing misleading positive interactions.

3. Global Min-Max Normalization

Since collaborative filtering scores lie on the 1–5 star scale and aspect scores lie on the [-1,1] range, both are normalized to $[0, 1]$ before blending:

$$ScoreCF_{\text{norm}}(u,i) = \frac{\text{clip}(S, c, oreCF(u,i), 1.0{,}5.0) - 1.0}{4.0}$$

$$ScoreAspect_{\text{norm}}(u,i) = \frac{ScoreAspect(u,i) + 1.0}{2.0}$$

If a user has no mapped training aspects, we set $ScoreAspect(u,i) = 0$, which yields a neutral normalized value of 0.5.

4. Final Blending

The final hybrid score is:

$$\hat{R}_{u,i} = \alpha \cdot ScoreCF_{\text{norm}}(u,i) + (1-\alpha) \cdot ScoreAspect_{\text{norm}}(u,i)$$

For evaluation, the blended score is rescaled back to the 1–5 rating range:

$$\hat{R}_{u,i}^{stars} = 1.0 + 4.0 \cdot \hat{R}_{u,i}$$

5. Early Stopping

To avoid overfitting, the matrix factorization model uses **early stopping** on the validation set. After each epoch, the model is evaluated on validation RMSE. Training stops if the validation score does not improve for a fixed number of consecutive epochs (patience), and the best-performing checkpoint is retained. This helps stabilize training and improves generalization.

6. CPU Optimization & Parallelization:

Explain how multiprocessing and batching in spaCy enabled processing under CPU constraints:

The full pipeline is designed to run under **CPU-only constraints**. Since spaCy-based text processing and review-level sentiment extraction are computationally expensive, documents are processed in batches to reduce overhead. Instead of parsing reviews one by one, the implementation uses:

$$\text{nlp.pipe}(\text{docs}, \text{batch_size}=1000, \text{n_process}=\text{K})$$

where K is a positive integer specified by the --n-process argument. This enables multiprocessing across independent documents and significantly improves throughput on large review collections.

A key implementation detail is that the **input order is preserved** when pairing processed documents with their corresponding review metadata. This is essential for correctly aligning extracted sentiment, aspect scores, and business/user identifiers after parallel parsing.

Parallelization is applied only to workloads that are independent across documents, such as tokenization, dependency parsing, and aspect-level sentiment extraction. This reduces wall-clock time without changing the model logic or the final recommendation scores.

This optimization is important for assignment-scale datasets because it keeps preprocessing and feature construction practical even without GPU acceleration.

7. Experimental Setup:

Hyperparameters (α, SVD latent dimensions k, learning rate, regularization, confidence threshold τ):

- alpha: blend weight between SVD and aspect-based scoring → 0.9
- k: latent dimension of the SVD model → 32
- learning rate: optimization step size for factorization training → 0.005
- regularization: penalty term to reduce overfitting and penalize large user/item biases and latent-factor magnitudes. This reduces overfitting by encouraging smoother parameter estimates and improving generalization on unseen user–item pairs. → 0.02
- Patience for early stopping → 5

$$\mathcal{L} = \sum_{(u,i)\in\mathcal{D}} (r_{ui} - \hat{r}_{ui})^2 + \lambda \left(b_u^2 + b_i^2 + \|\mathbf{p}_u\|^2 + \|\mathbf{q}_i\|^2\right)$$

- tau: threshold in hybrid recommendation fallback/abstention→ 0.3

8. Results & Comparison:

Metrics table comparing all 3 baselines against the Hybrid model (RMSE, HR@10, NDCG@10):

Model	RMSE	HR@10	NDCG@10
Popularity-based	1.055	0.076	0.0212
Rating-only SVD	1.005	0.029	0.0084
Aspect Content-Only	1.383	0.023	0.0076
Hybrid (alpha = 0.9)	1.005	0.025	0.0075

Table compares three baselines (Popularity-based, Rating-only SVD, Aspect Content-Only) with the Hybrid model ($\alpha = 0.9$) using RMSE, HR@10, and NDCG@10.

The Popularity-based baseline achieves the best top-10 ranking metrics (HR@10 = 0.076, NDCG@10 = 0.0212), despite a worse RMSE (1.055). This confirms that recommending globally popular items is highly effective for hit-rate and NDCG, even without personalization.

Rating-only SVD obtains the lowest RMSE (1.005), indicating the most accurate rating predictions, and moderate ranking performance (HR@10 = 0.029, NDCG@10 = 0.0084). In contrast, the Aspect Content-Only model performs worst overall (RMSE = 1.383, HR@10 = 0.023, NDCG@10 = 0.0076), showing that aspects and sentiment alone are too noisy to drive accurate recommendations.

The Hybrid model with $\alpha = 0.9$ matches Rating-only SVD in RMSE (1.005) but slightly underperforms it on ranking metrics (HR@10 = 0.025, NDCG@10 = 0.0075). This suggests that, in this configuration, injecting aspect information does not improve recommendation quality and may slightly weaken the pure collaborative signal. Overall, the experiment highlights a trade-off: **The results reveal a difference between rating prediction and top-**

k ranking objectives. Rating-only SVD achieves better RMSE because it is trained to predict users' numerical ratings. In contrast, the popularity-based model achieves higher HR@10 and NDCG@10 because it ranks globally popular businesses that are more likely to overlap with relevant items in the evaluation set. Therefore, popularity performs better for identifying relevant top-k items, whereas SVD is better calibrated for numerical rating prediction.

These results do not indicate that one model is universally better than the other. Instead, they show that model effectiveness depends on the evaluation objective: Rating-only SVD is superior for rating prediction, while the Popularity-based model is superior for top-k recommendation ranking.

9. Bonus Transformer Extensions (+30%): Document

1. Transformer designs:

Model Selection: We utilized DistilBERT, a distilled version of the BERT model that retains approximately 97% of BERT's performance while being 40% smaller and 60% faster. Specifically, the model fine-tuned on the SST-2 dataset was selected for its robust performance in binary sentiment classification. "distilbert-base-uncased-finetuned-sst-2-english" model works well on short sentences, tree structures and descriptive words (is fine tuned).

Scoring Logic: Unlike VADER (which is rule-based and uses a lexicon), this transformer model captures contextual nuances and word dependencies. The output label (POSITIVE/NEGATIVE) and its confidence score are mapped to a continuous range of [-1,1].

$$s = \begin{cases} +p & \text{if Label} = \text{"POSITIVE"} \\ -p & \text{if Label} = \text{"NEGATIVE"} \\ 0.0 & \text{otherwise} \end{cases}$$

2. implementation setup:

Pipeline: We integrated the model using the Hugging Face pipeline API for efficiency (does tokenization, padding, attention masking and forward pass automatically and fast)

Batching & Hardware: To optimize inference speed, we implemented batch processing (batch_size=16) and enabled CUDA acceleration (GPU) when available.

Integration: The transformer replaces the VADER sentiment component in the Aspect-Based Sentiment Analysis (ABSA) pipeline, providing more accurate sentiment weights for each extracted aspect and doesn't change other things like preprocessing.

3. computational efficiency:

VADER (Baseline): Near-instantaneous execution on CPU, extremely lightweight.

Transformer (Bonus): Higher computational overhead. Requires significant memory (VRAM/RAM) and benefits greatly from GPU acceleration. While slower than VADER, it provides a deeper semantic understanding of the reviews.

4. comparative evaluations against mandatory VADER (RMSE, HR@10, NDCG@10):

Model	RMSE	HR@10	NDCG@10
Hybrid (VADER) $(\alpha = 0.9)$	1.005	0.024	0.0075
Hybrid (Transformer) $(\alpha = 0.9)$	0.996	0.025	0.0081
Aspect Content-Only (VADER)	1.383	0.023	0.0076
Aspect Content-Only (Transformer)	1.086	0.023	0.0081
Rating-only SVD (VADER)	1.001	0.028	0.0085
Rating-only SVD (Transformer)	1.001	0.028	0.0085

The cost of time for transformer variant model was about 40239.96 seconds (about 11 hours), whereas for VADER was around 2 hours.

The empirical results in the table show a consistent improvement across all metrics when replacing the lexicon-based VADER sentiment analyzer with the Transformer-based model (DistilBERT).

- **Aspect Content-Only Sensitivity:** The impact of the Transformer model is most prominent in the *Aspect Content-Only* configuration. Since this baseline relies solely on the calculated user-aspect attention (U) and item-aspect quality (V) matrices, it is highly sensitive to the accuracy of the sentiment intensity values. Transitioning to the Transformer model leads to a significant reduction in RMSE from **1.383 to 1.086**. This suggests that VADER, being a rule-based lexicon scorer, struggles with contextual meanings, negation, and complex phrasing in reviews,

whereas the fine-tuned Transformer model captures these nuances far more effectively.

- **Hybrid Model Refinement:** In the *Hybrid model ($\alpha = 0.9$)*, where collaborative filtering (Biased MF) and content-based aspects are combined, the Transformer variant still yields the best overall performance (RMSE of **0.996**, HR@10 of **0.025**, and NDCG@10 of **0.0081**). While the rating prediction improvement is more modest here due to the regularizing effect of the collaborative filtering component, the ranking performance (NDCG@10) remains consistently superior.

10. Ablation Study:

At least 3 controlled experiments:

1. Alpha Sweeping:

Alpha	RMSE	HR@10	NDCG@10
0.0 (Aspect Content-Only)	1.383	0.023	0.0076
0.1	1.318	0.023	0.0078
0.2	1.258	0.023	0.0078
0.3	1.202	0.022	0.0075
0.4	1.153	0.024	0.0076
0.5	1.110	0.025	0.0078
0.6	1.072	0.025	0.0079
0.7	1.040	0.025	0.0077
0.8	1.018	0.023	0.0075
0.9	1.005	0.024	0.0075

1.0 (Ratings-Only SVD + fallback)	0.997	0.027	0.0083

The results show that increasing α generally improves RMSE. This indicates that the collaborative filtering component contributes more effectively to rating prediction than the aspect-based content component. Among the genuine hybrid configurations, $\alpha = 0.9$ achieves the lowest RMSE of 1.005, which is very close to the Ratings-Only SVD result of 0.997.

For ranking performance, $\alpha = 0.6$ achieves the highest NDCG@10 among the genuine hybrid configurations, with a value of 0.0079. Its HR@10 of 0.025 is also tied for the highest HR@10 among the hybrid configurations. Although $\alpha = 1.0$ obtains the best overall metrics, it is not a genuine hybrid model because the aspect component receives zero weight.

Therefore, $\alpha = 0.9$ **is selected when RMSE is the primary objective**, whereas $\alpha = 0.6$ **is preferable when ranking quality and a more balanced hybrid contribution are emphasized**. For the final hybrid model, $\alpha = 0.9$ can be justified as the best configuration for rating prediction because it retains a 10% contribution from the aspect-based component while achieving the lowest RMSE among all genuine hybrid settings.

2. Rating SVD alone vs. Hybrid (alpha = 0.9):

Model	RMSE	HR@10	NDCG@10
Rating-only SVD	1.005	0.029	0.0084
Hybrid (alpha = 0.9)	1.005	0.024	0.0075

Table compares Rating-only SVD with the Hybrid model ($\alpha = 0.9$). The two methods obtain the same RMSE (1.005), indicating no meaningful difference in rating prediction accuracy. However, Rating-only SVD performs better on ranking metrics, achieving higher HR@10 (0.029 vs. 0.024) and NDCG@10 (0.0084 vs. 0.0075). Overall, this shows that the hybrid combination at $\alpha = 0.9$does not improve top-k recommendation quality in this experiment.

3. Mandatory VADER vs. Transformer-based sentiment engine: (some explanations and table in part 9)

   Lexicon-based Approach: VADER

- Mechanism: VADER (Valence Aware Dictionary and sEntiment Reasoner) relies on a predefined gold-standard lexicon and a set of heuristic rules (e.g., handling negations like “not good” or intensifiers like “very good”).
- Limitations: While computationally efficient, VADER often fails to capture the contextual nuance of Yelp reviews. It processes words in isolation or short windows, leading to inaccurate sentiment scores for complex sentences or domain-specific terminology.
- Baseline Performance: Using VADER, the *Aspect Content-Only* model achieved an RMSE of 1.383. The higher error suggests that the sentiment scores provided by VADER were less representative of the actual star ratings assigned by users.

Deep Learning Approach: Transformer (DistilBERT)

- Mechanism: We integrated a fine-tuned DistilBERT model (distilbert-base-uncased-finetuned-sst-2-english). Unlike VADER, this model uses a **self-attention** mechanism to understand the relationship between words in a sequence, capturing bidirectional context.
- Improvements: The transition to the Transformer engine resulted in a significant performance boost:
- RMSE Reduction: In the *Aspect Content-Only* baseline, RMSE dropped from 1.383 to 1.086 (a ~21% improvement).
- Ranking Precision: Both HR@10 and NDCG@10 saw consistent gains, indicating that the Transformer accurately identified high-quality aspects that correlate better with user preferences.
- Complexity vs. Accuracy: Although the Transformer model requires more computational resources (GPU/inference time), the substantial reduction in error justifies its use in a hybrid recommendation setting where textual sentiment is a primary feature.

Infrastructure Note (The Fallback Effect)

It is important to note that the choice of the sentiment engine affects even the Ratings-Only SVD baseline reported in the final pipeline. Since all models share the same recommendation infrastructure and fallback logic, the improved precision of the Transformer-based sentiment scorer reduces the noise in the hybrid scoring stage, leading to more stable metrics across all variants compared to the VADER-based run.

11. Conceptual & Mathematical Answers:

Complete, detailed answers to the 3 questions in Section 4:

1) Global vs. local min-max normalization

Let $S_{CF}(u,\cdot) = \{s_1, \dots, s_n\}$ be the CF scores for all candidate items of user u, and let

$$s' = \frac{s - \min\big(S_{CF}(u,\cdot)\big)}{\max\big(S_{CF}(u,\cdot)\big) - \min\big(S_{CF}(u,\cdot)\big)}$$

be local min-max normalization. If $\mathrm{Var}\big(S_{CF}(u,\cdot)\big) = 0$, then all scores are identical, so

$$\max\big(S_{CF}(u,\cdot)\big) = \min\big(S_{CF}(u,\cdot)\big)$$

and the denominator becomes zero. Therefore local min-max normalization is undefined. The same argument applies to $S_{Aspect}(u,\cdot)$.

This is not just a numerical edge case. When the denominator is nearly zero, tiny floating-point perturbations can cause large swings in normalized values, which makes the ranking unstable. Two items with equal raw score can receive arbitrary different normalized scores depending on machine precision or sorting order. By contrast, global normalization uses fixed, user-independent bounds, so the denominator is constant and never collapses for a single user's candidate set. In code, this is exactly why CF is clipped to $[1, 5]$and then mapped to $[0, 1]$, while aspect scores are mapped from $[-, 11]$to $[0, 1]$. That gives a numerically stable scale even when a user's candidate scores are flat.

2) Review bias and sentiment calibration

User ratings and textual sentiment are related but not identical. A review like "the service was okay, but the pasta was legendary" can contain mixed sentiment, while the user may still assign 5 stars due to overall satisfaction, price, context, or personal bias. So the rating distribution and the VADER compound-score distribution are not on the same latent scale.

Mathematically, if the aspect alignment score is computed from sentiment-weighted user preferences, then a mismatch between rating scale and sentiment scale introduces bias into the score

$$S_{Aspect}(u,i) = \sum_{a} U\,[u,a]V[i,a]$$

because U is estimated from sentiment evidence that may be systematically shifted relative to star ratings. If rating inflation is common, then U may overstate positive aspects even when the text is only mildly positive, causing overconfident aspect scores. If the opposite happens, aspect scores are undercalibrated.

A practical calibration strategy is to clip and rescale both signals to compatible ranges before blending. In this project, the aspect score is clipped to [-1,1] and then transformed to $[0, 1]$:

$$S_{Aspect}^{norm}(u, i) = \frac{S_{Aspect}(u, i) + 1}{2}$$

For ratings, the equivalent global normalization is:

$$S_{CF}^{norm}(u, i) = \frac{\text{clip}(S_{CF}, (u, i), 15) - 1}{4}$$

A stronger calibration variant is to use monotone quantile normalization or z-score followed by clipping, but the fixed-range mapping above is simpler and fully consistent with your implementation.

3) Profile matrix sparsity bound

Let $U \in \mathbb{R}^{|U| \times 5}$be the user-aspect preference matrix. Each user reviews at most $R_{\max}$items, and each review mentions at most $A_{\max} \leq 5$ unique aspects. Then each user can have at most

$$K = R_{\max} A_{\max}$$

aspect mentions in total, but since there are only 5 aspect columns, the number of nonzero entries per user is at most

$$\min(5, K)$$

Hence the number of zero entries per user is at least

$$5 - \min(5, K)$$

and the sparsity ratio per user is at least

$$\frac{5 - \min(5, R_{\max} A_{\max})}{5}.$$

So:

- If $R_{\max}A_{\max} < 5$, a non-trivial lower bound exists:

$$\text{sparsity} \geq 1 - \frac{R_{\max}A_{\max}}{5}.$$

- If $R_{\max}A_{\max} \geq 5$, the bound becomes trivial:

$$\text{sparsity} \geq 0.$$

The role of $|U|$ is that it scales the total matrix size, but the per-user sparsity bound is independent of $|U|$. Across the full matrix, the total number of entries is $5|U|$, and the total number of nonzeros is at most $|U| \min(5, R_{\max}A_{\max})$, giving the same sparsity ratio after division. So $|U|$ affects absolute storage size, not the lower bound on sparsity percentage.

Project implementation already matches this structure: U is built as a |users| × 5 DataFrame, and each row is normalized by the total mentioned aspects, so sparsity is controlled by how many aspects a user ever mentions, not by the number of users overall.

12. Error Analysis:

Examine 5 concrete failure cases:

1. ≥ 2 aspect noise/sarcasm (in dev_predictions_hybrid.csv):
   Text1:
   review_id: 1lUQygYCJwVYFV8IVnz9bA
   user_id: HhjX7i0NkqBsiA2k6STojw
   business_id: XsfuqDqw9eOaQ_2FCv5MuQ
   stars: 2.0
   predicted_stars: 4.041758791056286
   score_aspect_norm: 0.5
   text: Damn. Not an option to try. Few things management should consider. First is aloo kathi roll is literally aloo curry and roti where are flavors of chutney??? Second, I tried kala khatta and it's bull$hit please remove it from the menu.
   Why it fails: Words like “damn”,” literally”,” bull$hit” are sarcastic and aspect noise.

   Text2:
   review_id: 1ugMVWNT-acfySd4NcxkCg
   user_id: f1hAlR_mJsv8Gd1CSNRYpg
   business_id: C809UuprygJyEgJw4wr2Pg

stars: 1.0
predicted_stars: 2.0221059613701886
score_aspect_norm: 0.5
text: Apparently this is sweet mango habanero! And I even ordered the sauce on the side but where is it? If I had wanted just plain mediocre wings I would've fried some myself instead of pay 6.99 for some shit wing dings. And the guy on the phone had the audacity to hang up on me
Why it fails: sarcastic sentences like “Apparently this is sweet mango habanero! And I even ordered the sauce on the side but where is it?” and aspect noise like “mediocre wings”, “shit wing dings”,” audacity to hang up”

2. ≥ 2 rating/aspect conflict (in dev_predictions_hybrid.csv):
   Text1:
   review_id: 5D0ST8VmXAOH-drDnf5YXQ
   user_id: lahQZzrrOG9xxGiqs977iQ
   business_id: zPl2SR6S2-X4T40DQdnoTw
   stars: 4.0
   predicted_stars: 2.5834643380684357
   score_aspect_norm: 0.5303398064321846
   text: The mozzarella sticks are good , the fries are too even tho they came with no ketchup or anything, the cheesesteak is smaller than expected & not very flavorful but it's not bad.
   Why it fails: mixed-sentiment/ambiguous
   - Positive aspects: good, fries are too
   - Negative aspects: no ketchup, smaller than expected, not very flavorful
   - Mixed signals: even tho, not bad

   Text2:

   review_id: TnqxmdGvakyZYpBvrvFPFQ
   user_id: o0n5f395W9HSRZRhbps0hQ
   business_id: 0RuvlgTnKFbX3IK0ZOOocA
   stars: 5.0
   predicted_stars: 3.1619957921293524
   score_aspect_norm: 0.5891033176389495

   text: NOT OFFERING DINE IN, even tho online it says they are, not a big deal tho.

   Also NOT OFFERING VEGAN SAUSAGE. That bothered me a bit but it isn't on the menu so I can't complain, I did ask because that's what my stomach was set on. Customer

service was acceptable, Food was great as usual, over this is a positive review just wanted people to know it's no dine or Vegan Sausage lol

Why it fails: mixed-sentiment/ambiguous

- Positive aspects: great, positive review
- Negative aspects: bothered, NOT OFFERING DINE IN, NOT OFFERING VEGAN SAUSAGE
- Mixed signals: not a big deal tho, I can't complain, a bit

3. ≥ 1 cold-start user (in dev_recommendations.csv):
   user_id: -3h48rG-sopyGDLu5bh4xQ
   This user is a fallback case: all top-10 recommendations are marked is_fallback=True. This suggests the system could not rely on normal personalized CF scoring and used fallback recommendations instead. The file does not reveal the exact reason, but it is consistent with a cold-start or sparse-history user.
   The same thing for **user_id:** -Gi7vsHoui-W6e_-SYh9kA.

13. AI Usage Disclosure:

Transparent description of AI tools used during development:

AI tools were used during development to assist with debugging, code explanation, report structuring, and refinement of technical wording. Final implementation decisions, experimental validation, and result interpretation were performed by the author.